

phantom Function Test Results

Test Case 1 : Default Phantom

"Image Size:" 256. 256.

"Image Sum:" 8044.0000

"Mean Intensity:" 0.1227417

"Output:"

1.	0.69	0.92	0.	0.	0.
-0.8	0.6624	0.874	0.	-0.0184	0.
-0.2	0.11	0.31	0.22	0.	-18.
-0.2	0.16	0.41	-0.22	0.	18.
0.1	0.21	0.25	0.	0.35	0.
0.1	0.046	0.046	0.	0.1	0.
0.1	0.046	0.046	0.	-0.1	0.
0.1	0.046	0.023	-0.08	-0.605	0.
0.1	0.023	0.023	0.	-0.606	0.
0.1	0.023	0.046	0.06	-0.605	0.

Test Case 2 : Small Image (N = 8)

"Image Size:" 8. 8.

"Image Sum:" 7.6000000

"Output:"

0.	0.	0.	0.	0.	0.	0.	0.
0.	0.	1.	0.2	0.2	1.	0.	0.
0.	0.	0.2	0.3	0.3	0.2	0.	0.
0.	0.	0.2	-5.551D-17	0.2	0.2	0.	0.

0.	0.	0.2	-5.551D-17	-5.551D-17	0.2	0.	0.
0.	0.	0.2	0.2	0.2	0.2	0.	0.
0.	0.	1.	0.2	0.2	1.	0.	0.
0.	0.	0.	0.	0.	0.	0.	0.

Test Case 3 : 90 Degree Rotation

"Image Size:" 128. 128.

"Image Sum:" 1400.

"Center Pixel:" 1.

"Non-zero Pixels:" 1400.

Test Case 4 : Shepp-Logan Phantom

"Image Size:" 256. 256.

"Minimum Intensity:" 0.

"Maximum Intensity:" 1.

"Output:"

1.	0.69	0.92	0.	0.	0.
-0.98	0.6624	0.874	0.	-0.0184	0.
-0.02	0.11	0.31	0.22	0.	-18.
-0.02	0.16	0.41	-0.22	0.	18.
0.01	0.21	0.25	0.	0.35	0.
0.01	0.046	0.046	0.	0.1	0.
0.01	0.046	0.046	0.	-0.1	0.
0.01	0.046	0.023	-0.08	-0.605	0.
0.01	0.023	0.023	0.	-0.606	0.
0.01	0.023	0.046	0.06	-0.605	0.

Test Case 5 : Negative Intensity Ellipse

"Image Size:" 128. 128.

"Minimum Intensity:" -0.8

"Maximum Intensity:" 0.

"Image Sum:" -1212.8000

Test Case 6 : Ellipse Partially Outside Image

"Image Size:" 128. 128.

"Image Sum:" 1240.

"Non-zero Pixels:" 1240.

Test Case 7 : Rotated and Shifted Ellipses

"Image Size:" 256. 256.

"Image Sum:" 2169.

"Center Pixel:" 0.

Test Case 8 : Overlapping Ellipses

"Image Size:" 101. 101.

"Maximum Intensity:" 1.7

"Non-zero Pixels:" 1255.

"Image Sum:" 2133.5000

Test Case 9 : Ellipses Completely Outside Image

"Image Size:" 128. 128.

"Image Sum:" 0.

"Non-zero Pixels:" 0.

Test Case 10 : Invalid Inputs

"Case 1 : Invalid Model"

"ERROR" "phantom: unknown MODEL"

"Case 2 : Invalid Ellipse Matrix"

"ERROR" "phantom: first argument must either be MODEL, E, or N"

"Case 3 : Invalid Numeric Input"

"ERROR" "phantom: first argument must either be MODEL, E, or N"

"Case 4 : Invalid Second Argument"

"ERROR" "phantom: N must be numeric scalar"

"Case 5 : Too Many Inputs"

"ERROR" "phantom: Wrong number of input arguments."